

Calculus A

Chapter 2: Limits and Continuity

Section 2.5: Continuity

Main Lecture Notes

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Reference: Thomas' Calculus, 12th Edition

Continuity

Fact

For a function to be continuous at a point, it must be defined at that point, its limit must exist at the point, and the value of the function at that point must equal the value of the limit at that point.

Definition: Continuity at a point

A function $f(x)$ is **continuous at a point** a if and only if the following three conditions are satisfied:

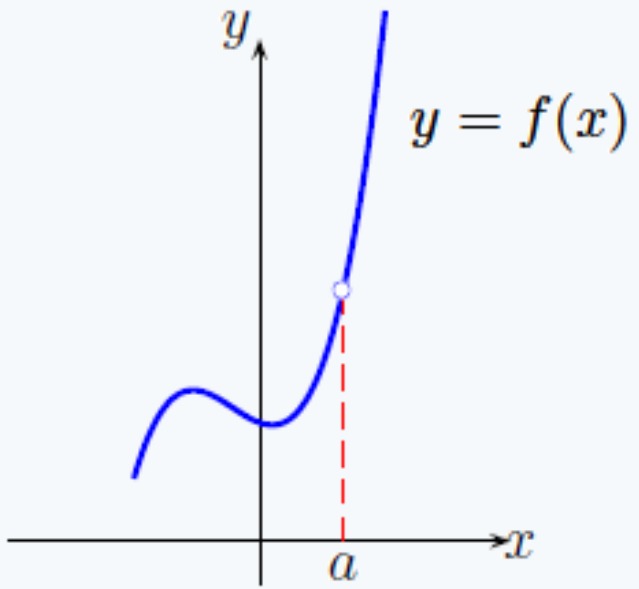
1. $f(a)$ is defined. 2. $\lim_{x \to a} f(x)$ exists. 3. $\lim_{x \to a} f(x) = f(a)$

Definition: Continuity at a point Cont..

A function f fails to be continuous at a if one or more of the following conditions holds:

1. If f is not defined at a.
2. If $lim_{x\to a} f(x)$ does not exist.
3. $lim_{x\to a} f(x)$ exist but it is not equal to $f(a)$.

If one of the above conditions holds we say f is **discontinuous at** a. See the types of dicontinuity in the next slide.

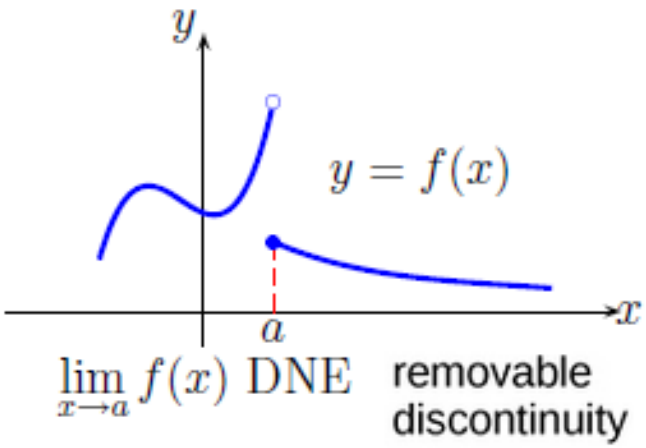


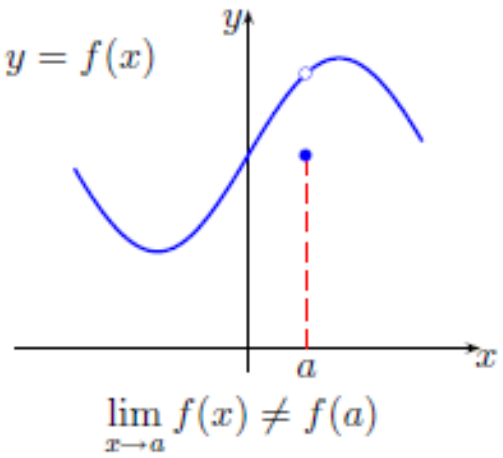
$f(a)$ is not defined

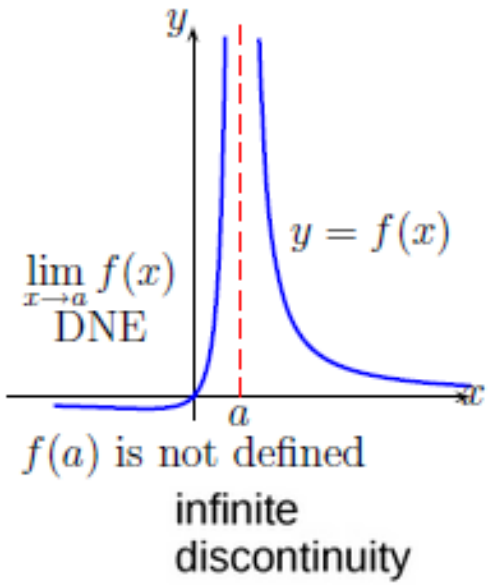
Continuity and Discontinuity

Types of Discontinuity

We classify the types of discontinuities as **removable** discontinuities, **infinite** discontinuities, or **jump** discontinuities. Intuitively, a removable discontinuity is a discontinuity for which there is a hole in the graph, a jump discontinuity is a non-infinite discontinuity for which the sections of the function do not meet up, and an infinite discontinuity is a discontinuity located at infinity.







Example 1

Determine whether the function

$$h(x) = \begin{cases} \dfrac{x^2+2x-3}{x-1}, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

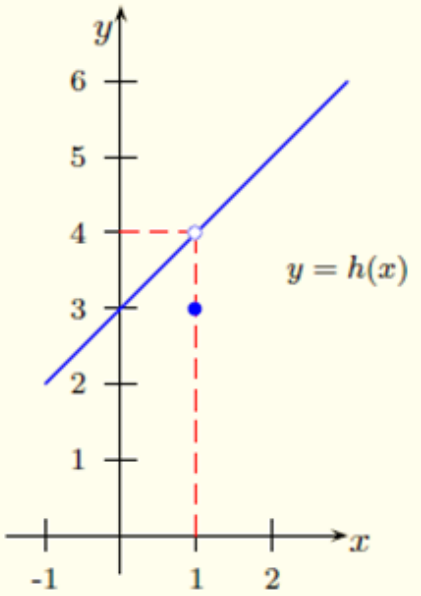
is continuous at $a = 1$.

Solution:

The function h is defined at $a = 1$ with $h(1) = 3$.

$$\begin{aligned} \lim_{x\to 1} h(x) &= \lim_{x\to 1} \frac{x^2+2x-3}{x-1} \\ &= \lim_{x\to 1} \frac{\cancel{(x-1)}(x+3)}{\cancel{(x-1)}} \\ &= \lim_{x\to 1} (x+3) = 4. \end{aligned}$$

Now, since $\lim_{x\to 1} h(x) = 4 \neq 3 = h(1)$, then h is discontinuous at $a = 1$.



Continuity and Discontinuity

Example 2

The graph of f is given to the right. At which numbers is f discontinuous? Justify your answer.

Solution:

The function f is discontinuous at $x = -2$ because $f(-2)$ is not defined and

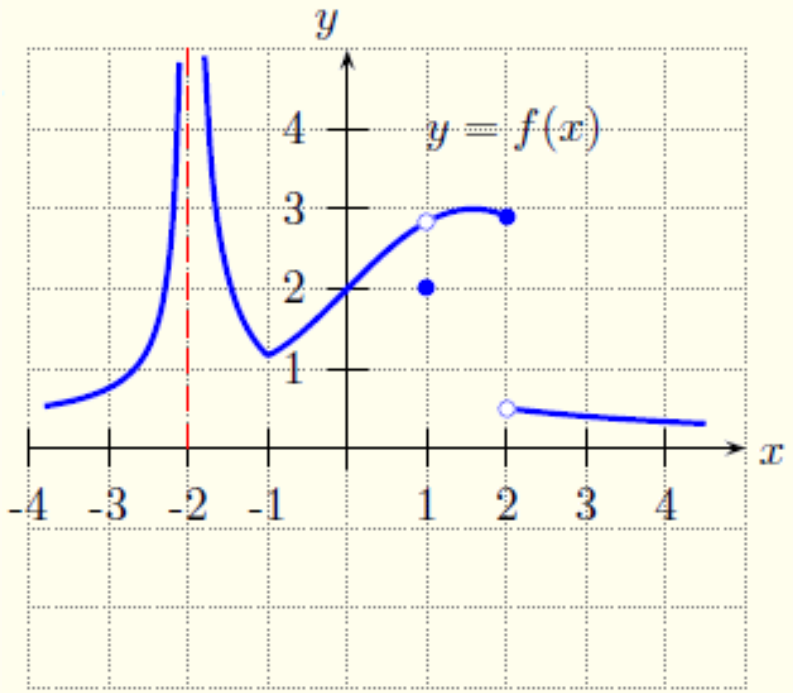
$$\lim_{x \to -2} f(x) \quad \text{DNE.}$$

The function f is discontinuous at $x = 1$ because

$$f(1) = 2 \neq 2.8 \approx \lim_{x \to 1} f(x).$$

The function f is discontinuous at $x = 2$ because

$$\lim_{x \to 2^+} f(x) \approx 0.5 \neq 2.9 \approx \lim_{x \to 2^-} f(x).$$



Note: At every other point c in the domain $\mathcal{D}_f$, the function $f(x)$ is continuous.

One-Sided Continuity and Continuity on a Closed Interval

Definition

Interior point: A function $y = f(x)$ is **continuous at an interior point** c of its domain if

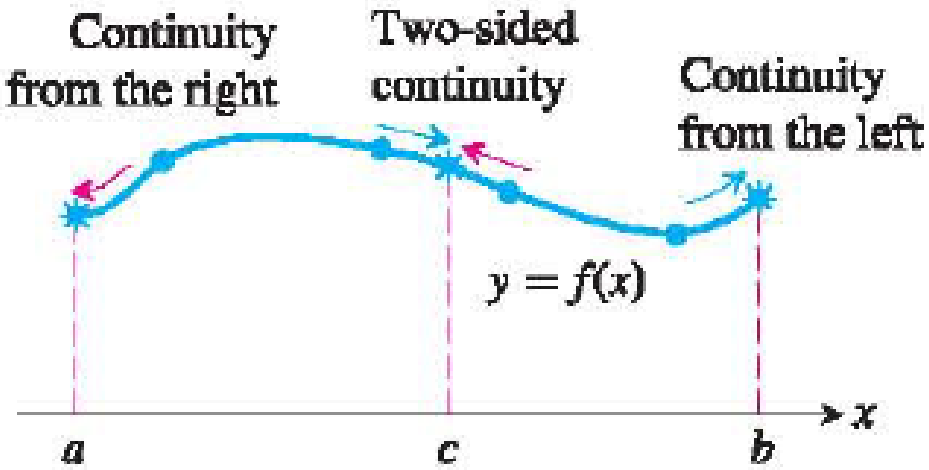
$$\lim_{x \to c} f(x) = f(c).$$

Endpoint: A function $y = f(x)$ is **continuous at a left endpoint** a if

$$\lim_{x \to a^+} f(x) = f(a).$$

A function $y = f(x)$ is **continuous at a right endpoint** b if

$$\lim_{x \to b^-} f(x) = f(b).$$



Continuity at points a, b, and c.

One-Sided Continuity

Example 3

Discuss the continuity of the integer functions $\lfloor x \rfloor$, (the greatest integer function) and $\lceil x \rceil$, (the least integer function).

Solution:

The greatest integer function, $f(x) = \lfloor x \rfloor$ is discontinuous at every integer because the left-hand and right-hand limits are not equal

$$\lim_{x \to n^-} \lfloor x \rfloor = n - 1 \quad \text{and} \quad \lim_{x \to n^+} \lfloor x \rfloor = n$$

Since $\lfloor n \rfloor = n$, the greatest integer function is right-continuous at every integer n (but not left-continuous).

Example 3 Cont..

Discuss the continuity of the integer functions $\lfloor x \rfloor$, (the greatest integer function) and $\lceil x \rceil$, (the least integer function).

Solution Cont..

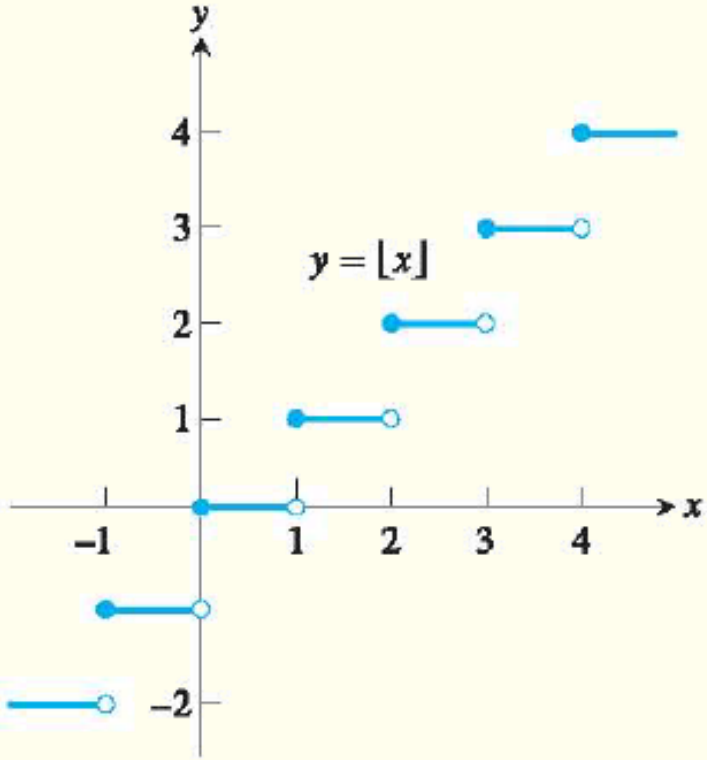
The greatest integer function is continuous at every real number other than the integers. For example,

$$\lim_{x \to 1.5} \lfloor x \rfloor = 1 = \lfloor 1.5 \rfloor.$$

In general, if $n - 1 < c < n$, n an integer, then

$$\lim_{x \to c} \lfloor x \rfloor = n - 1 = \lfloor c \rfloor.$$

H.W.: Use the same method to discuss the continuity of the least integer function, $\lceil x \rceil$.



Continuity on a Closed Interval

Definition

A function is continuous over an open interval (a, b) if it is continuous at every point in the interval.

A function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b) and is continuous from the right at a i.e.,

$$\lim_{x \to a^+} f(x) = f(a),$$

and is continuous from the left at b i.e.,

$$\lim_{x \to b^-} f(x) = f(b).$$

Continuity on a Closed Interval

Example 4

Discuss the continuity of $f(x) = \sqrt{25 - x^2}$.

Solution:

Notice that the domain of f is the closed interval $\mathcal{D}_f = [-5, 5]$, so we need check the continuity of f on the open interval $(-5, 5)$ and at the endpoints. Let $a \in (-5, 5)$, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} \sqrt{25 - x^2} = \sqrt{25 - a^2} = f(a)$$

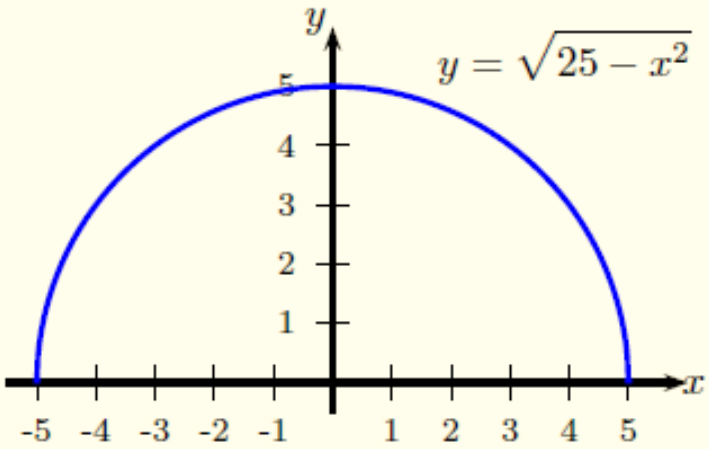
Hence f is continuous on the open interval $(-5, 5)$. Also f is continuous at the endpoints because

$$\lim_{x \to -5^+} f(x) = \lim_{x \to -5^+} \sqrt{25 - x^2} = 0 = f(-5)$$

and

$$\lim_{x \to 5^-} f(x) = \lim_{x \to 5^-} \sqrt{25 - x^2} = 0 = f(5).$$

Therefore, f is continuous on the closed interval $[-5, 5]$.



Continuity on Intervals

Example 5

Determine the intervals on which $f(x) = \dfrac{x^2 + x - 12}{x^2 - 3x}$ is continuous.

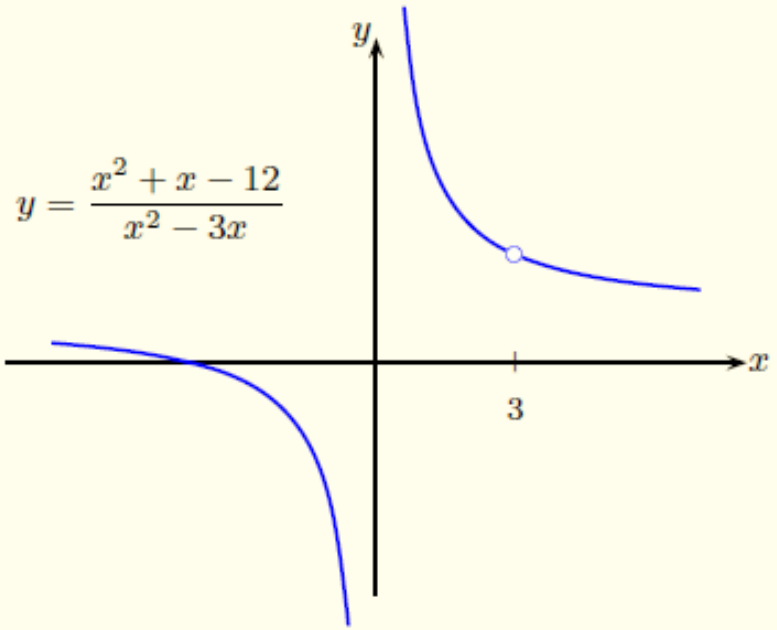
Solution:

Notice that $f(x) = \dfrac{x^2 + x - 12}{x^2 - 3x}$ is a rational function, then it is continuous on its domain which is the set of real number except the zeroes of the denominator. Now

$$\begin{aligned} x^2 - 3x = 0 &\Leftrightarrow x(x-3) = 0 \\ &\Leftrightarrow x = 0 \text{ or } x = 3. \end{aligned}$$

Hence f is continuous on

$$\mathbb{R}\backslash\{0, 3\} = (-\infty, 0) \cup (0, 3) \cup (3, +\infty).$$



Continuity on Intervals

Example 6

Find all discontinuities of

$$g(x) = \begin{cases} 2x+1, & x \le 0, \\ \dfrac{\sqrt{x+1}-1}{x}, & x > 0. \end{cases}$$

Solution:

Notice that $g(x)$ is continuous for any $x \neq 0$. For $x = 0$, we have $g(0) = 2(0)+1 = 1$. To find $\lim_{x\to 0} g(x)$ we have to compute $\lim_{x\to 0^+} g(x)$ and $\lim_{x\to 0^-} g(x)$.

$$\lim_{x\to 0^-} g(x) = \lim_{x\to 0^-} \frac{\sqrt{x+1}-1}{x}$$ substitution will give I.F. $\left(\frac{0}{0}\right)$

$$= \lim_{x\to 0^-} \frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}$$ use $(a-b)(a+b) = a^2 - b^2$.

Continuity on Intervals

Example 6 Cont..

$$g(x) = \begin{cases} 2x+1, & x \le 0, \\ \dfrac{\sqrt{x+1}-1}{x}, & x > 0. \end{cases}$$

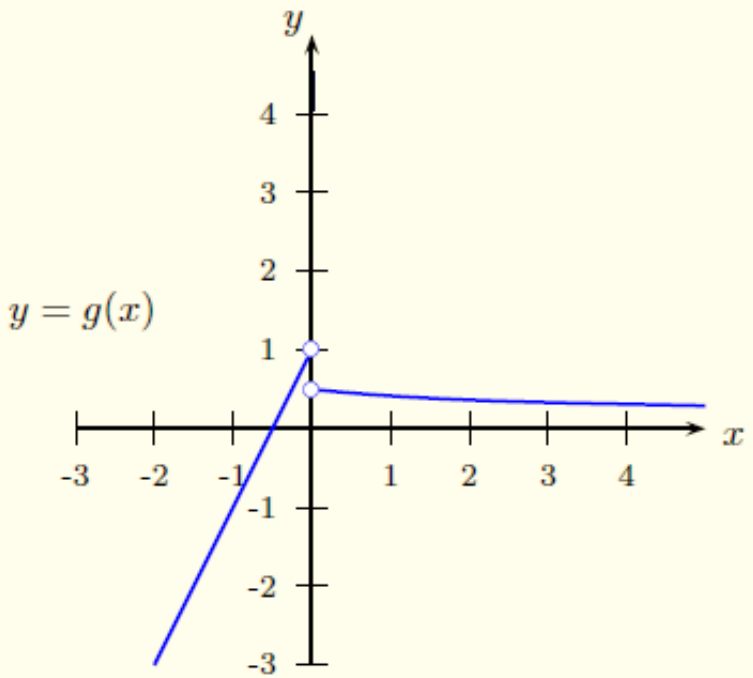
Solution Cont..

$$\begin{aligned} \lim_{x\to 0^-} g(x) &= \lim_{x\to 0^-} \frac{(x+1)-1}{x(\sqrt{x+1}+1)} \\ &= \lim_{x\to 0^-} \frac{\cancel{x}}{\cancel{x}(\sqrt{x+1}+1)} \\ &= \lim_{x\to 0^-} \frac{1}{\sqrt{x+1}+1} = \frac{1}{2} \end{aligned}$$

While,

$$\lim_{x\to 0^+} g(x) = \lim_{x\to 0^+} (2x+1) = 1.$$

Since $\lim_{x\to 0^-} g(x) = \frac{1}{2} \neq 1 = \lim_{x\to 0^+} g(x)$. Thus $\lim_{x\to 0} g(x)$ DNE.
Hence g is discontinuous at $x = 0$.



Continuity on Intervals

Example 7

Find the value of a that makes the given function continuous on $\mathbb{R}$.

$$f(x) = \begin{cases} 5x - 2, & x \geq 2, \\ ax^2 + 2, & x < 2. \end{cases}$$

Solution:

f is continuous on the set of real numbers if it is continuous at $x = 2$, and f is continuous at $x = 2$ if and only if $\lim\limits_{x \to 0^+} f(x) = \lim\limits_{x \to 0^-} f(x)$.

Now,

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (5x - 2) = 8 \quad \text{and}$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (ax^2 + 2) = 4a + 2$$

Hence

$$4a + 2 = 8 \iff 4a = 6 \iff a = \frac{6}{4} = \frac{3}{2}.$$

Continuity on Intervals

Example 8

State the interval(s) over which the function

$$f(x) = \frac{x-1}{x^2+2x}$$

is continuous.

Solution:

Since $f(x) = \frac{x-1}{x^2+2x}$ is a rational function, it is continuous at every point in its domain. The domain of $f(x)$ is the set $(-\infty, -2) \cup (-2, 0) \cup (0, +\infty)$. Thus, $f(x)$ is continuous over each of the intervals $(-\infty, -2)$, $(-2, 0)$, and $(0, +\infty)$.

Example 9

State the interval over which the function $f(x) = \sqrt{x+3}$ is continuous.

Solution:

$f(x)$ is continuous over the interval $[-3, +\infty)$. (Check !!!)

Properties of Continuous Functions

THEOREM 8—Properties of Continuous Functions If the functions f and g are continuous at $x = c$, then the following combinations are continuous at $x = c$.

1.	*Sums:*	$f + g$
2.	*Differences:*	$f - g$
3.	*Constant multiples:*	$k \cdot f$, for any number k
4.	*Products:*	$f \cdot g$
5.	*Quotients:*	f/g, provided $g(c) \neq 0$
6.	*Powers:*	f^n, n a positive integer
7.	*Roots:*	$\sqrt[n]{f}$, provided it is defined on an open interval containing c, where n is a positive integer

Properties of Continuous Functions

Theorem: Composite of Continuous Functions

If $f(x)$ is continuous at L and $lim_{x\to a}g(x) = L$, then

$$lim_{x\to a}f(g(x)) = f(lim_{x\to a}g(x)) = f(L).$$

Example 10: Evaluate

$$(a)\ \lim_{x\to\frac{\pi}{2}}\cos\left(x-\frac{\pi}{2}\right),\ (b)\ \lim_{x\to\pi}\sin(x-\pi),\ \text{and}\ (c)\ \lim_{x\to 0}\sin\left(\frac{\pi+x}{x+1}\right)$$

Solution:

The given functions are a composite of $\cos x$ or $\sin x$ and polynomials and since all are continuous everywhere, we may apply the composite function theorem. Thus,

$$(a)\ \lim_{x\to\frac{\pi}{2}}\cos\left(x-\frac{\pi}{2}\right) = \cos\left(\lim_{x\to\frac{\pi}{2}}\left(x-\frac{\pi}{2}\right)\right) = \cos(0) = 1,$$

$$(b)\ \lim_{x\to\pi}\sin(x-\pi) = \sin(\lim_{x\to\pi}(x-\pi)) = \sin(\pi-\pi) = \sin 0 = 0,\quad \text{and}$$

$$(c)\ \lim_{x\to 0}\sin\left(\frac{\pi+x}{x+1}\right) = \sin\left(\lim_{x\to 0}\left(\frac{\pi+x}{x+1}\right)\right) = \sin(\pi) = 0.$$

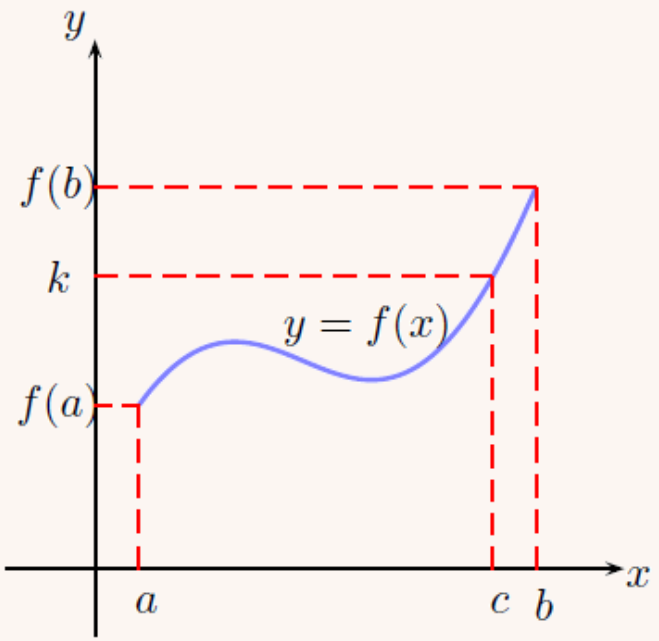
The Intermediate Value Theorem

Theorem

If f is continuous on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a, b]$ such that $f(c) = k$.

Fact

The intermediate value theorem states that as x varies from a to b, the continuous function f takes on every value between $f(a)$ and $f(b)$. If we think of a continuous function as a function whose graph has no hole or break extending from the point $(a, f(a))$ to the point $(b, f(b))$, as illustrated in the Figure, then for any number k between $f(a)$ and $f(b)$, the horizontal line $y = k$ intersects the graph in at least at one point. The $x-$coordinate c of the intersection point is the number such that $f(c) = k$.



The Intermediate Value Theorem

Example 11

Show that function $f(x) = x^3 + 2x^2 - 1$ has a zero in the interval $[0, 1]$.

Solution:

Notice that $f(x) = x^3 + 2x^2 - 1$ is polynomial function, then it is continuous on its domain which is the set of real number. Hence $f(x) = x^3 + 2x^2 - 1$ is continuous on the closed interval $[0, 1]$. Now, since

$$f(0) = (0)^3 + 2(0)^2 - 1 = -1 \quad \text{and}$$
$$f(1) = (1)^3 + 2(1)^2 - 1 = 1 + 2 - 1 = 2.$$

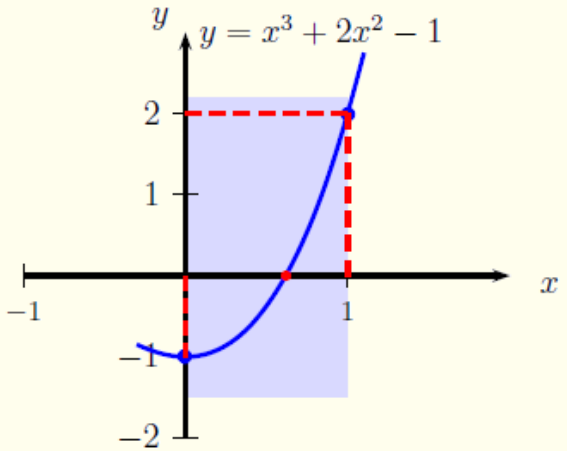
Hence

$$f(0) = -1 < 0 < 2 = f(1).$$

Therefore by the Intermediate Value Theorem there exist some c in $[0, 1]$ such that $f(c) = 0$. Hence

$$f(x) = x^3 + 2x^2 - 1$$

has a zero in the interval $[0, 1]$.



Using The Intermediate Value Theorem

Example 12

Use the Intermediate Value Theorem to show that there is a solution to the equation $x^2 = \sqrt{x+2}$ in the interval $[-2, 4]$.

Solution:

We can look to the equation as follows;

$$x^2 = \sqrt{x+2} \Rightarrow \sqrt{x+2} - x^2 = 0 \quad \text{So, let } f(x) = \sqrt{x+2} - x^2$$

Since $\sqrt{x+2}$ is a root function, it is continuous in its domain, $x \geq -2$. Also, since x^2 is a polynomial, it is continuous on its domain $\mathbb{R} = (-\infty, \infty)$. Now the difference of two continuous functions is continuous. So f is continuous on its domain $\{x \in \mathbb{R} : x \geq -2\} = [-2, \infty)$, thus f is continuous over $[-2, 4]$. Now,

$$f(-2) = \sqrt{-2+2} - (-2)^2 = -4 \quad \text{and}$$
$$f(0) = \sqrt{0+2} - (0)^2 = \sqrt{2}.$$

Since $f(-2) < 0$ and $f(0) > 0$, by the Intermediate Value Theorem there is a number $c \in [-2, 4]$ such that $f(c) = 0$.

End of Section 2.5.

Thank you.!!!